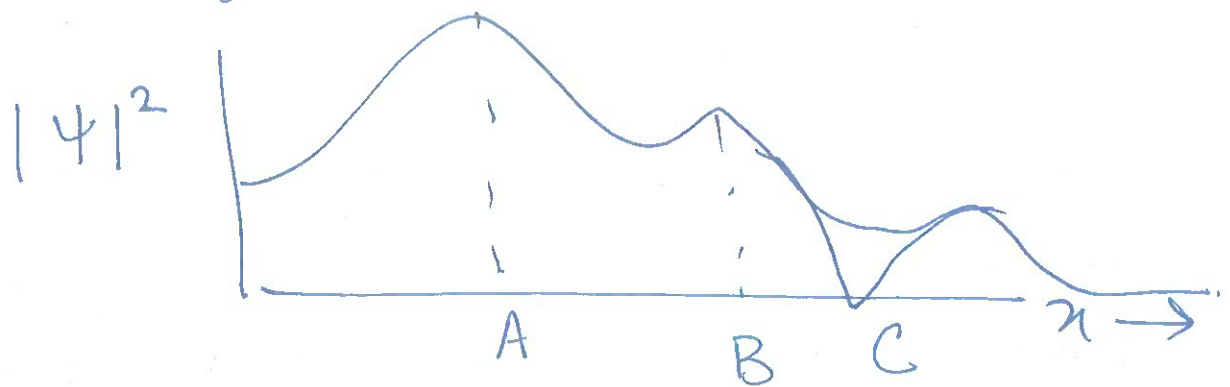


Schrodinger equation

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi$$



$|\psi|^2$ defines the probability density or the probability of finding the particle in the vicinity of A, B, C points. let's say.

- 2) from the plot the particle is likely to be found at A and less likely to be found at B compared to A and can not be found at C.
- 3) QM can not give exact position of particle rather provides the statistical prediction about the possible position — it only gives us probability associated with the particle's existence.

4) In case we make a measurement and find the particle at B — the question arises where was the particle just before the measurement.

5) The most sensible answer will be that it was at B.

Then QM is not a complete theory since it could not tell us the same with 100% certainty.

6) The other possible answer would be it was at any other point but the measurement forced the particle to be at B.

7) Hence the observation or measurement not only disturbs what it is to ^{be} measured but it also produces the result.

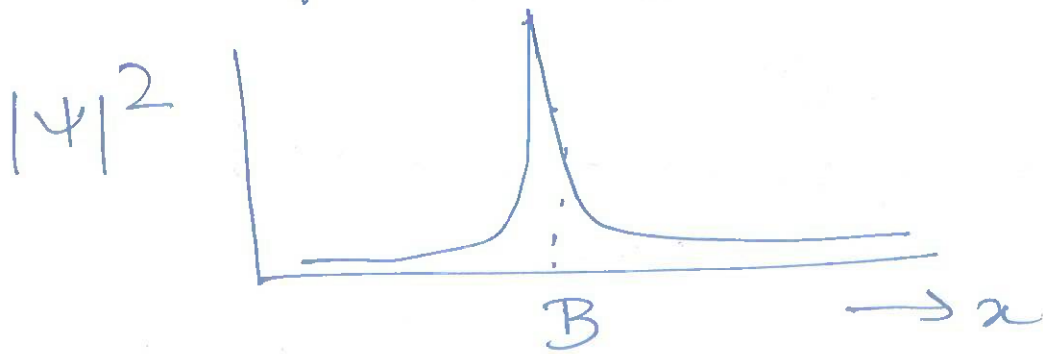
— the first view was advocated by Einstein and his followers.

— the 2nd ~~of~~ view was represented by Niels Bohr and followers

— this is the most accepted view of QM.

$\frac{1}{2}$ So the measurement force the particle to be at B — if we make a new measurement what will be the outcome.

- a) It was agreed upon that the 2nd measurement will also detect the particle at B — the measurement forces the wavefn. to peak at B.



- 10) The wavefn. quickly spread over according to S.E hence the 2nd measurement must be done quickly.

- 11) Two aspects : ① Measurement —→ collapse of wavefn.
② Evolution —→ spread of " according to S.E

properties of ψ and $|\psi|^2$

- 1) ψ is usually complex.
- 2) probability must be real quantity b/w 0 - 1.

3) $\int_a^b |\psi|^2 dx = 1$
called normalization

Condⁿ.

~~Acc~~ by Acceptable Wavefⁿ. Can be made normalized.

How do we get Schrödinger eqn?

① Let us consider a free particle ($F=0$) moving in the $+x$ direction

A free particle is represented as a plane wave $y(x,t) = A e^{-i\omega(t - \frac{x}{v})}$ — (i)

is The equation to represent motion of particle.

v = velocity of the particle.

x = displacement.

$$\omega = 2\pi\nu$$

$$v = \lambda\nu$$

$$\omega = 2\pi\nu \quad \text{angular velocity} \quad \left. \vphantom{\omega = 2\pi\nu} \right\} \text{for fundamental Particle}$$

$$\nu = \frac{1}{\lambda}$$

hence

$$\psi(x,t) = A e^{-i(2\pi\nu)\left(t - \frac{x}{\lambda\nu}\right)}$$

$$= A e^{-2\pi i\left(\nu t - \frac{x}{\lambda}\right)} \quad \left[\frac{1}{\lambda} = k \right] \text{ wave number}$$

Now $E = h\nu = 2\pi\hbar\nu$

$$\lambda = \frac{h}{p} = \frac{2\pi\hbar}{p}$$

hence

$$\psi = A e^{-i2\pi\left(\frac{Et}{2\pi\hbar} - \frac{xp}{2\pi\hbar}\right)}$$

$$= A e^{-i\left(Et/\hbar - xp/\hbar\right)} \quad \text{---(ii)}$$

Now $\psi \rightarrow \Psi$ for quantum particles.

$$\Psi = A e^{-i(Et/\hbar - xp/\hbar)} \quad \text{---(iii)}$$

$$\Psi(x,t) = A e^{i xp/\hbar} e^{-i Et/\hbar}$$

this is the eqn of motion for a free particle in QM.

(2) So $\Psi = A e^{-\frac{i}{\hbar}(Et - px)}$

Represent equation of motion of a free particle — since we are talking about quantum particle the eqn. contains $\hbar$.

(3) This eqn. describes a particle with energy E and momentum moving in direction x — unrestricted particle.

(4) In order to arrive at S.E

$$i\hbar \frac{\partial \Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \Psi}{\partial x^2} + V\Psi$$

Let us differentiate eqn. (iii) wrt 't'

$$\begin{aligned} (5) \quad \frac{\partial \Psi}{\partial t} &= A e^{-\frac{i}{\hbar}(Et - px)} \cdot \frac{-iE}{\hbar} A e^{-\frac{i}{\hbar}(Et - px)} \\ &= -\frac{iE}{\hbar} \Psi \quad \dots (iv) \end{aligned}$$

(6) Now if we double differentiate wrt x .

$$\frac{\partial \psi}{\partial x} = A \frac{ip}{\hbar} e^{-(i/\hbar)(Et - xp)}$$

$$= A \left(\frac{ip}{\hbar} \right) e^{-(i/\hbar)(Et - xp)}$$

$$= \left(\frac{ip}{\hbar} \right) \psi$$

$$\frac{\partial^2 \psi}{\partial x^2} = \left(\frac{ip}{\hbar} \right) \left(\frac{ip}{\hbar} \right) \psi = -\frac{p^2}{\hbar^2} \psi$$

$$= -\frac{p^2}{\hbar^2} \psi$$

hence $p^2 \psi = -\hbar^2 \frac{\partial^2 \psi}{\partial x^2}$

$$\underline{p^2 \psi} = \left(i\hbar \frac{\partial}{\partial x} \right) \left(i\hbar \frac{\partial}{\partial x} \right) \psi$$

double differential of space —

single differentiation of ψ wrt x

gives momentum

$$p = mv = m x/t$$

$$\begin{aligned} \text{K.E} &= \frac{1}{2} mv^2 \\ &= \frac{1}{2} m \left(\frac{x^2}{t^2} \right) \end{aligned}$$

⑦ According to S.E

$i\hbar \frac{\partial}{\partial t}$ corresponds to energy E (observable)

while $-i\hbar \frac{\partial}{\partial x}$ corresponds to momentum p (observable)

$i\hbar \frac{\partial}{\partial t} \equiv$ total energy operator

~~this is also called~~
energy operator causes time translation.

⑧ We represent operators with 'caret'

$\hat{p} = -i\hbar \frac{\partial}{\partial x}$ momentum operator

$\hat{E} = i\hbar \frac{\partial}{\partial t}$ Energy operator

⑨ Classical energy $\underline{T.E} = K.E + P.E$

$$E = \frac{p^2}{2m} + V \Rightarrow i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2 \psi}{\partial x^2} + V\psi$$

$$\frac{1}{2m} v^2 \quad K.E = \frac{1}{2} m v^2 = \frac{1}{2} (mv) v = \frac{p \cdot p}{2m} = \frac{p^2}{2m}$$

Interpretation of S.E

$$i\hbar \frac{\partial}{\partial t} \Psi(x, t) = \hat{H} \Psi(x, t).$$

- 1) A linear ~~diff~~ partial differential eqn.
- 2) First order in time
- 3) Second order in space.
- 4) deterministic in the evolution of Ψ
it allows us to calculate $\Psi(x, t)$
at a $t = t$ when $\Psi(x, 0)$ is known.
provided $\hat{H}$
- 5) ~~Give~~ An initial wavefn $\Psi(x, 0)$
and a Hamiltonian $\hat{H}$
will give us $\Psi(x, t)$ at all future
time. | Total energy generates time translat.

Physical interpretation: Ψ is not observable

$|\Psi|^2$ = probability of
finding the particle
at x at t .

So. S.E tells us how probability
amplitude evolves in time.

Hence S.E is the QM analog of classical energy expression.

(10) Operators

$$\text{P.E} \rightarrow V(x, t)$$

$$\text{K.E} \rightarrow -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2}$$

$$\hat{p} \Rightarrow -i\hbar \frac{\partial}{\partial x}$$

$$\text{T.E } \hat{E} \rightarrow i\hbar \frac{\partial}{\partial t}$$

$$\text{position } x \rightarrow \hat{x}$$

$$(11) \quad i\hbar \frac{\partial \psi}{\partial t} = \cancel{\frac{p^2}{2m}} - \frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} \psi + V\psi$$

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi$$

$\hat{H}$ = called the Hamiltonian operator or energy operator.

(12) Energy operator generates time translation or time evolution

operators

momentum operator

→ spatial translation

angular momentum

→ rotations

Energy → time translation

(13)

In physics

Continuous symmetries are generated by operator.

For. eg. if we consider that laws of physics does not change over time if we move from $t \rightarrow t+a$

— This sym. implies there exists an operator $\hat{U}(a)$ such that

$$\psi(t+a) = \hat{U}(a)\psi(t)$$

Since time translation forms a continuous group $\hat{U}(a)$ must be of exp. form $\hat{U}(a) = e^{-i/\hbar a \hat{H}}$

Hence Here, $\hat{H}$ by definition is the generator of time evolution

(14) for infinitesimal time evolution

$$\psi(t+dt) \approx \left(1 - \frac{i}{\hbar} \hat{H} dt\right) \psi(t)$$

it can be rearranged to

$$\boxed{i\hbar \frac{d\psi}{dt} = \hat{H} \psi(t)}$$

this is what is Schrödinger equation.

— S.E is the definition of what it means for ~~these~~ energy to generate time evolution.

(15) Physical meaning of Hamiltonian

$\hat{H}$ — is not just a number called "energy"

it is 1) an operator that tells ~~us~~ the state how to change in time.

2) A quantum version of the classical energy pr.

$$\hat{H} \psi_E = E \psi_E$$

time evolution $\psi_E(t) = e^{-\frac{iEt}{\hbar}} \psi(0)$

(16) Energy determines how fast the phase rotates in time.

For

— Stationary States ~~stay fixed~~

i) probability distribution stay fixed. while

ii) Their phase evolves

(17) $\hbar$ High energy $\rightarrow$ faster phase rotation
High momentum $\rightarrow$ faster Spatial translation

————— X ————— X —————

Definition of Hamiltonian

Hamiltonian is the operator whose exponential shift-quantum state forward in time, just as momentum shift states in space.

Properties of Wavefn.

Condⁿ. for acceptable wavefn.

- ① wavefn. must be single valued
For any given ~~time~~ value of x & t
the wavefn. $\psi(x, t)$ must have
a unique value.

This is a way to ~~generalize~~ guarantee that there is only a
single value for the prob. of
the system being in a given
state.

- ② must be square integrable
③ must be continuous So that 1st-order derivative can be defined.
④ All first order derivatives must be continuous. So that 2nd order derivative can be defined

1. e^{-x} $(0, \infty)$ acceptable

single valued, continuous
finite, quadratically integrable

2. e^{-x} $(-\infty, \infty)$ not acceptable

3. $4e^{x^2}$ $(-\infty, \infty)$ not acceptable.

